



Project Roadmap: ECG-Based Heart Disease Classification System

Role: Senior Data Scientist, Cardiology Informatics Lab

Goal: Build, evaluate, and interpret multiple ML models (Logistic Regression, Decision Tree, Random Forest, KNN, XGBoost) to identify the best classifier for predicting cardiovascular risk using ECG and clinical indicators.

[Erbil Heart Disease Dataset](#)

PHASE 1 — Data Understanding & Clinical Context

Data Dictionary

Column Name	Data Type	Description	Possible Values / Units	Clinical Meaning
age	Numeric (int)	Patient age at the time of assessment	Years	Age is a primary risk factor for cardiovascular disease; risk increases with age.
sex	Binary (0/1)	Biological sex of the patient	0 = Female, 1 = Male	Male patients often show higher early cardiovascular risk.
smoke	Binary (0/1)	Current smoking status	0 = Non-smoker, 1 = Smoker	Smoking is a major modifiable cardiovascular risk factor.
years	Numeric (float/int)	Total years of smoking	Years	Indicates cumulative smoking exposure (pack-years proxy).

fh	Binary (0/1)	Family history of cardiovascular disease	0 = No, 1 = Yes	Family history increases inherited or genetic heart disease risk.
ldl	Numeric	LDL cholesterol level	mg/dL (typical clinical units unless otherwise specified)	Elevated LDL is strongly associated with atherosclerosis and ischemic heart disease.
chp	Categorical (1–4)	Cholesterol phenotype or class	1–4 (encoded)	Represents clinical lipid categories; higher values may indicate more abnormal lipid profiles.
height	Numeric	Patient height	cm	Used for BMI and body surface area calculations.
weight	Numeric	Patient weight	kg	Used to compute BMI; obesity is a known cardiovascular risk factor.
bpsys	Numeric	Systolic blood pressure	mmHg	The upper BP value; elevated systolic pressure is linked to hypertension and stroke risk.
bpdias	Numeric	Diastolic blood pressure	mmHg	The lower BP value; used with systolic BP to evaluate overall BP control.
hr	Numeric	Resting heart rate	beats per minute (bpm)	Abnormal HR (too low or too high) can indicate cardiac conduction issues or heart disease.
ihd	Binary (0/1)	Diagnosed Ischemic Heart Disease	0 = No, 1 = Yes	Indicates existing coronary artery disease; may affect ECG patterns.
dm	Binary (0/1)	Diabetes mellitus status	0 = No, 1 = Yes	Diabetes is a strong cardiovascular risk multiplier.

htn	Binary (0/1)	Hypertension status	0 = No, 1 = Yes	High blood pressure significantly increases cardiac disease risk.
active	Binary (0/1)	Physical activity status	0 = Inactive, 1 = Active	Low activity levels increase metabolic and cardiac risks.
ivsd	Numeric	Interventricular septal thickness	mm	Thickening may indicate left ventricular hypertrophy or cardiac remodeling.
ecgpatt	Categorical (1–4)	ECG pattern category	1,2,3,4 (encoded)	Represents types of ECG waveform patterns; may indicate ischemia or conduction abnormalities.
qwave	Binary (0/1)	Presence of pathological Q-wave on ECG	0 = Absent, 1 = Present	Pathological Q-waves often indicate previous myocardial infarction.
target	Binary (0/1)	Disease outcome / cardiovascular risk classification	0 = Negative (No disease), 1 = Positive (Disease present)	The dependent variable used in model training.

Task 1: Define the clinical objective

- Understand from cardiologists what the “target” represents (presence/absence of disease).
- Clarify what actionable insights the predictive model must provide to clinicians.

Task 2: Initial dataset inspection

- Load dataset and document:
 - Number of rows and columns

- Data types
 - Missing values
 - Unique counts for categorical variables
 - Validate that all features correspond to clinical meaning (ldl, hr, qwave, ecg pattern, hypertension, diabetes, etc.)
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PHASE 2 — In-Depth Exploratory Data Analysis (EDA)

Task 3: Univariate analysis

For each numerical feature:

- Measure central tendencies and dispersion
- Check skewness, kurtosis
- Plot histograms & boxplots
- Identify physiologically impossible values

For each categorical/binary:

- Generate frequency plots
- Check for class imbalance

Task 4: Bivariate analysis (feature → target relationships)

- Compare distributions of each numeric feature across target classes
- Study ECG pattern contribution to disease occurrence

- Assess lifestyle factors (smoking, activity, family history → disease outcomes)

Task 5: Multivariate analysis

- Compute correlations among numeric features
 - Generate Pearson heatmap
 - Assess redundancy or multicollinearity
 - Identify clinically significant clusters (e.g., cholesterol + hypertension + diabetes)
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PHASE 3 — Statistical Testing & Hypothesis Validation

As a hospital data scientist, ensure statistical rigor to validate clinical relevance.

Task 6: Continuous vs Target (binary)

- Use independent samples t-tests
- Use point-biserial correlations

Objective: Identify numerical biomarkers strongly associated with disease.

Task 7: Categorical vs Target

- Use Chi-Square test of independence

Objective: Validate ECG patterns, hypertension, diabetes, smoking as risk contributors.

Task 8: Continuous vs Categorical (>2 groups)

- Perform One-Way ANOVA
- **Objective:** See whether clinical measurements differ significantly across ECG pattern types.

Task 9: Continuous vs Continuous

- Pearson correlation for linear relationships
 - Spearman correlation for monotonic relationships
Objective: Detect dependencies such as LDL ↔ blood pressure.
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PHASE 4 — Data Cleaning & Transformation

Task 10: Handle missing values

- Document missingness percentage
- Apply medically appropriate imputations for:
 - Lab values
 - Vitals
 - ECG interpretations

Task 11: Outlier detection & treatment

- Identify outliers using:

- IQR method
- Visual outlier check using boxplots
- Decide whether to:
 - Cap extreme values
 - Remove physiologically impossible points
 - Leave medically plausible extremes that reflect real pathology

Task 12: Data transformation

- Apply Yeo-Johnson transformation to skewed numeric variables
 - Standardize continuous features
 - Encode categorical and binary predictors properly
- Objective:** Ensure all models receive clean, normalized, properly-scaled data.
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PHASE 5 — Feature Engineering & Selection

Task 13: Create useful clinical features

- Derive BMI from height and weight
- Create risk interaction features (e.g., LDL × Diabetes)
- Categorize age groups (if needed)
- Combine systolic & diastolic BP to compute pulse pressure

Task 14: Feature importance screening

Use:

- Statistical tests
 - Correlation results
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PHASE 6 — Model Development (Pipeline-Driven)

Task 15: Split data

- Stratify by target to preserve class ratios
- Train/validation/test split

Task 16: Build models

For each model:

1. Logistic Regression
 2. Decision Tree
 3. Random Forest
 4. K-Nearest Neighbors
 5. XGBoost
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PHASE 7 — Model Evaluation & Selection

Task 17: Evaluate models on test set using:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC
- AUC (Area Under ROC Curve)
- Confusion matrix

Task 19: Choose best model

Determine:

- Best performing model

Document all results in a comparison table.

PHASE 8 — Model Interpretation & Clinical Explainability

Task 20: Measure feature importance

Explain to cardiologists:

- Which factors drive predictions

- How much each contributes
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PHASE 9 — Reporting & Deployment Preparation

Prepare deployment handoff

- Provide final cleaned dataset
- Save the model using Pickle and Provide trained model files